Palo Alto Networks PAN-OS — Administrator Password Recovery

Zero-Data-Loss Standard Operating Procedure: Leveraging Maintenance Mode ('maint') & config.xml.backup to Restore 100% of Production Policies with a New Administrative Credential

Document ID:	SOP-PAN-REC-01	**Appliance Models:**	PA-220 / 400 / 800 / 3200 / 5200 / PA-VM
Target OS:	PAN-OS 9.1 / 10.0 / 10.1 / 10.2 / 11.0+	**Console Baud / Port:**	9600-8-N-1 (Physical) / TCP 5008 (GNS3)
Classification:	Operational Runbook — Enterprise Security	**Lead Engineer:**	Nishu Saini (Innspark Solutions)

1. Operational Context: The Forgotten Administrator Password Dilemma

In enterprise network engineering, losing access to a production Palo Alto Next-Generation Firewall presents severe business risks. Unlike commercial consumer routers or standard web portals, **Palo Alto PAN-OS intentionally excludes any 'Forgot Password' button or web-based recovery mechanism** on its HTTPS management login page. This security architecture ensures that an external or unauthorized adversary cannot compromise administrative rights over the management plane.

Historically, engineers faced with a forgotten password might resort to a full factory reset (request system private-data-reset). However, a factory reset completely purges all configured Security Policies, NAT rules, Decryption profiles, Site-to-Site IPsec tunnels, GlobalProtect VPN certificates, and Layer 3 interfaces. This document details the **Zero-Data-Loss Disaster Recovery Workflow**: a method that safely swaps the administrative credentials while recovering **100% of the latest running configuration**.

2. Architectural Mechanism: How the 'config.xml.backup' Safety Net Works

The core technical mechanism that makes zero-data-loss recovery possible is built into the PAN-OS Maintenance Mode storage engine. When an engineer boots into Maintenance Mode (maint) and restores an older configuration XML file (where credentials are known), the PAN-OS underlying Linux kernel automatically executes a pre-commit preservation hook: it takes the current active production running-config.xml and saves an exact snapshot of it to local flash storage named:

```
config.xml.backup
```

Because config.xml.backup contains every single policy, zone, address object, and rule that was active before the lockout, we can log in using our old credentials, load config.xml.backup into the **Candidate Configuration** buffer, immediately overwrite the admin password in GUI before committing, and execute a commit. The result is total recovery without data loss.

Phase	Operational Action	Storage State & Mechanics
Phase 1: Boot 'maint'	Reboot firewall via console; enter **maint** at prompt.	Bypasses active PAN-OS daemons into RAM-disk recovery kernel.
Phase 2: Restore Old Config	Select an older backup XML with a known password. Reboot.	PAN-OS saves current config as **config.xml.backup**.
Phase 3: Web GUI Login	Open https://<MGT-IP> and log in with known old credentials.	Firewall runs old baseline config; full GUI access is restored.
Phase 4: Load Latest Backup	Navigate to **Device** → **Setup** → **Operations** → **Load named configuration snapshot**. Select config.xml.backup.	Loads all latest policies/NAT into **Candidate Config** buffer.
Phase 5: Task Audit	Open bottom-right **Tasks** panel. Verify Config Load is **Completed**.	Confirms candidate buffer is 100% loaded and syntax valid.
Phase 6: Password Override	Navigate to **Device** → **Administrators** → **admin** → **Change Password**. Set new secure password.	Overwrites the forgotten password inside candidate buffer **BEFORE commit**.
Phase 7: Final Commit	Click **Commit** at top-right. Commit job reaches 100% success.	Candidate becomes Running Config. Latest setup is active with new password.

3. Step-by-Step Tactical Execution Guide

Follow these seven exact phases in strict chronological sequence. Do not skip or alter the order of operations:

STEP 1: Serial Console Access & Maintenance Mode Boot

1. Attach to the serial console port of the firewall (**RJ-45 9600-8-N-1** on physical hardware, or port **5008** in GNS3/EVE-NG).
2. Reboot the firewall appliance (power-cycle or send break sequence).
3. As the kernel initializes, observe the serial output. When the countdown prompt appears:
   Hit Enter to continue...
   Immediately type **maint** and press **Enter** within 5 seconds.

STEP 2: Load Known Configuration in 'maint' Mode & Trigger Auto-Backup

1. The PAN-OS Maintenance Recovery Tool loads into a blue/gray text-based terminal interface.
2. Press **Enter** on **Continue**.
3. Navigate using arrow keys to **Config Recovery** (or **Restore Configuration**).
4. Select an older saved configuration XML file where the username and password are known.
5. Confirm the restore. During this operation, PAN-OS copies the active running configuration to config.xml.backup.
6. Select **Reboot** to restart the firewall into normal operational mode.

STEP 3: Web GUI Authentication with Known Old Credentials

1. Allow the firewall 3–5 minutes to complete daemon startup (Management Plane, mgmtsrvr, and Web Server).
2. Open a web browser (Google Chrome or Mozilla Firefox) and navigate to the management interface:
   https://172.17.0.50 (or your OOB management IP e.g. https://192.168.1.1).
3. Bypass the self-signed SSL certificate prompt (**Advanced** → **Proceed to IP (unsafe)**).
4. Log in using the **OLD known credentials** from the snapshot restored in Phase 2.

STEP 4: Load 'config.xml.backup' into Candidate Configuration Buffer

1. In the top navigation bar of the Web GUI, click the **Device** tab.
2. In the left-hand navigation pane, select **Setup** → **Operations**.
3. Locate the **Configuration Management** section.
4. Click **Load named configuration snapshot**.
5. In the modal dialog dropdown, locate and select the auto-saved backup file:
   config.xml.backup
6. Click **OK**.

[CRITICAL WARNING]: DO NOT click the 'Commit' button at this point! Clicking commit now will lock you out again with the forgotten password.

STEP 5: Verify Job Completion in PAN-OS Task Manager

1. Look at the bottom-right corner of the PAN-OS Web GUI footer bar.
2. Click the **Tasks** button (icon with clipboard/checkboxes) to open the Task Manager window.
3. Locate the job labeled **Config Load**.
4. Verify that the *Status* column displays a green checkmark and says **Completed**, with *Result* showing **FIN** (100%).
5. Click **Close** to dismiss the Task Manager.

STEP 6: Pre-Commit Administrator Password Reset

1. Navigate to **Device** → **Administrators** in the left navigation tree.
2. In the Administrator Accounts table, click directly on the username **admin**.
3. In the *Administrator* pop-up configuration dialog, click the **Change Password** link / checkbox.
4. Type your desired **New Password** (ensure it satisfies corporate complexity: ≥ 8 characters, upper, lower, numbers, and symbols).
5. Type the identical string into **Confirm Password**.
6. Click **OK** to close the administrator dialog.

[CRITICAL WARNING]: This action updates the admin password inside the Candidate Configuration buffer BEFORE it is committed to production.

STEP 7: Execute Permanent Commit & Authentication Validation

1. Click the blue **Commit** button located at the extreme top-right corner of the Web GUI screen.
2. In the *Commit* verification window, click **Commit**.
3. Monitor the commit progress bar as PAN-OS compiles and validates the configuration.
4. Once the commit job reaches **100% (Success)**, close the dialog.
5. **Verification:** Log out of the Web GUI and verify that you can authenticate successfully using **admin** and your **NEW password**.

4. Alternative PAN-OS CLI Commands (For Headless / Terminal Engineers)

For network engineers operating strictly via SSH or serial console after Phase 3, the exact equivalent CLI commands to perform Steps 4 through 7 are:

```
# 1. Enter Configuration Mode:
admin@PA-VM> configure

# 2. Load the automatically saved production backup into candidate buffer:
admin@PA-VM# load config from config.xml.backup

# 3. Immediately overwrite the admin password in the candidate buffer (DO NOT COMMIT YET):
admin@PA-VM# set mgt-config users admin password
Enter password:  <TypeYourNewPassword123!>
Confirm password: <TypeYourNewPassword123!>

# 4. Commit candidate configuration to activate all policies and the new password:
admin@PA-VM# commit

# 5. Exit configuration mode and verify active sessions and administrators:
admin@PA-VM# exit
admin@PA-VM> show admins all
admin@PA-VM> show system info
```

5. Critical Pitfalls & Operational Safety Matrix

Failure Condition	Underlying Cause	Immediate Remediation Action
Immediate Re-Lockout	Engineer clicked **Commit** right after loading config.xml.backup without changing password first.	Re-execute from Phase 1. Maintenance Mode will still preserve your latest configuration as a backup.
Commit Failed (Complexity)	New password does not meet PAN-OS minimum complexity requirements (length, character classes).	Ensure password has at least 8 characters, 1 uppercase, 1 lowercase, 1 digit, and 1 special character.
Web GUI Unreachable after Reboot	Old restored snapshot had a different management IP (e.g. 192.168.1.1 instead of 172.17.0.50).	Connect via console, check IP with show interface management, or re-assign with set deviceconfig system ip-address ....
Missed 'maint' Prompt	Failed to type maint within 5-second countdown on serial console.	Reboot firewall again via console. Keep cursor active and type maint repeatedly as kernel initializes.

ENGINEERING CERTIFICATION & COMPLIANCE

Zero-Data-Loss Guarantee: By executing this runbook, no Security Policies, NAT rules, Zones, or VPN tunnels are discarded. The PAN-OS engine seamlessly transitions from the emergency recovery configuration back to full enterprise production status.